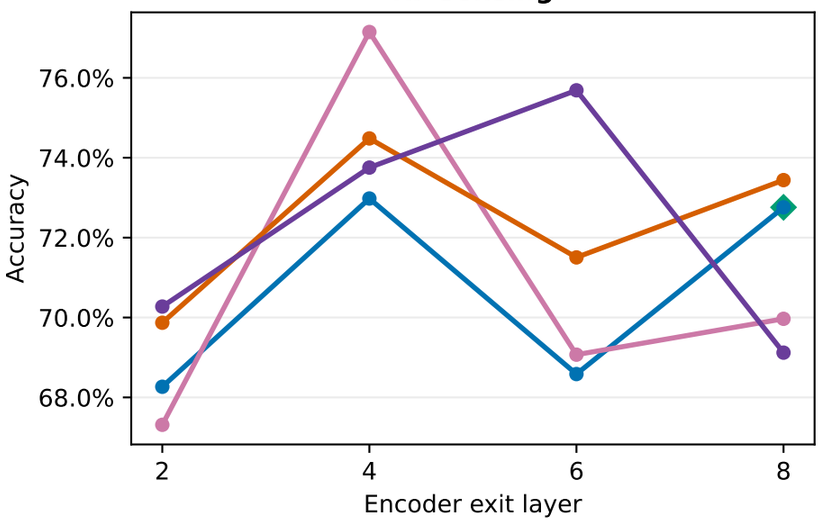
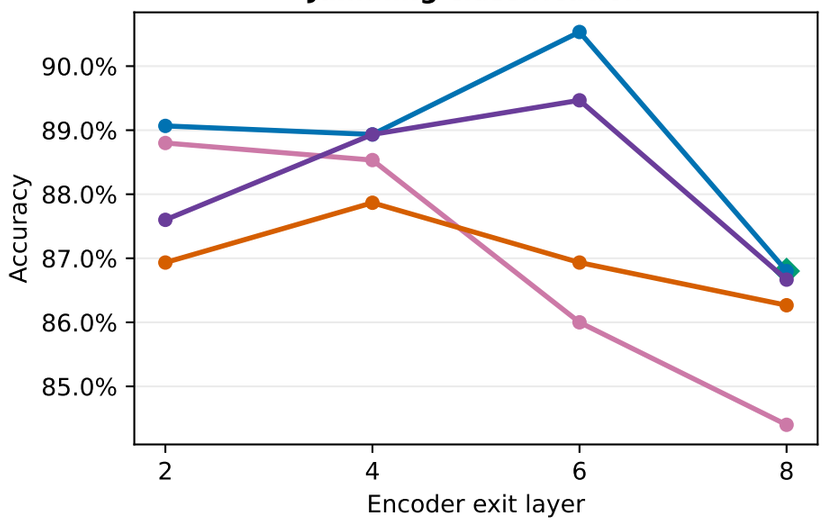
Downstream performance across the approaches

B0: Original LatentWave (full-depth)
P0: LatentWave + post-hoc frozen exits
P1: Same-scale multi-depth JEPA
P2: Proposed hierarchical JEPA
P2-Reverse: Coarse-to-fine hierarchical JEPA

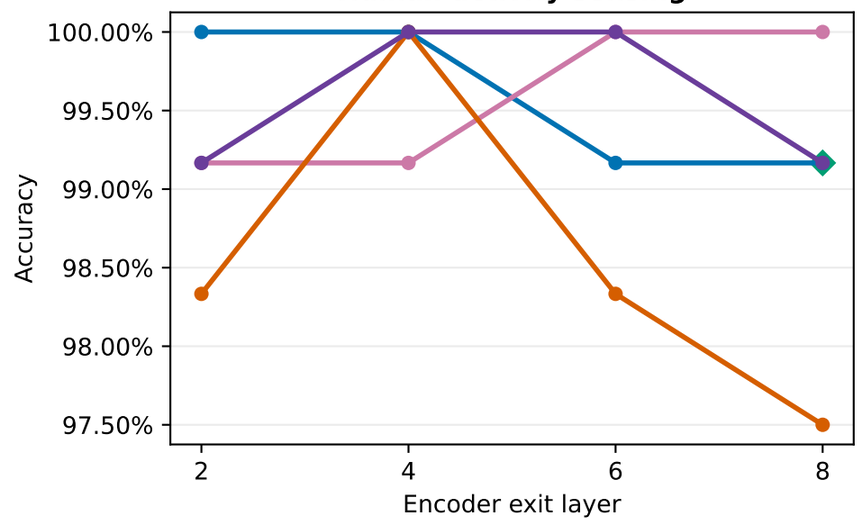
RF sensing



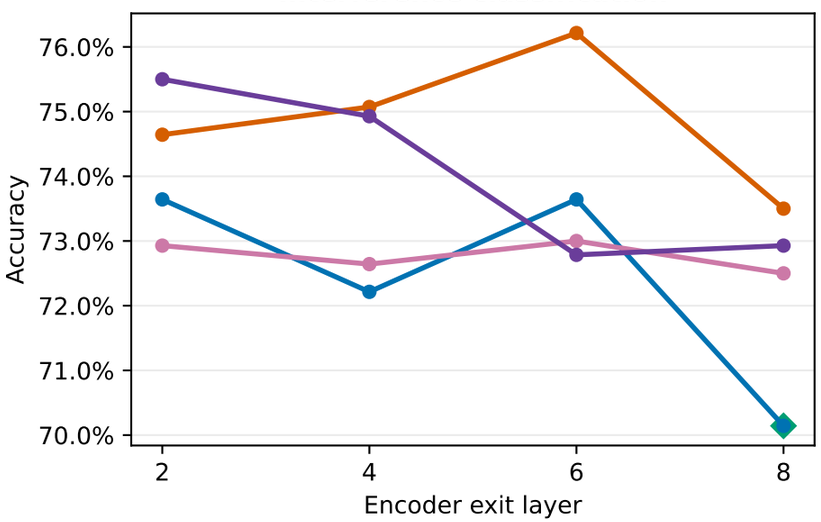
Jamming classification



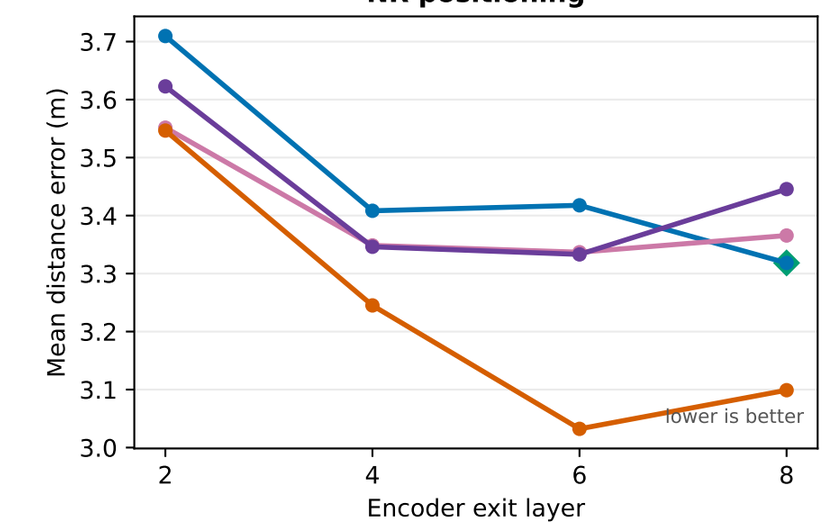
Human activity sensing



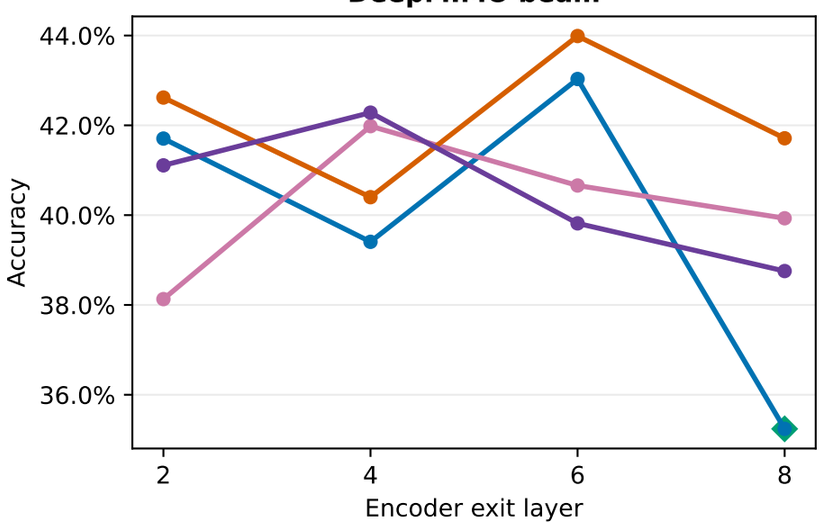
Interference classification



NR positioning



DeepMIMO beam



Diamond: Original LatentWave at full-depth L8. Circles: frozen-head exits. All rows use the shared 80/10/10 split and frozen context encoders.